

KAMALESH KUMAR

857-919-6625 • kamalesh.kumar@unh.edu • linkedin.com/in/kamalesh1806 • STEM OPT Eligible

EDUCATION

Masters in Business Analytics

Aug 2025 - Sep 2026

University of New Hampshire, Durham, NH

Bachelor of Computer Science

Aug 2020 - April 2023

Guru Nanak College Of Arts and Science

GPA: 3.58

WORK EXPERIENCE

Associate software engineer, Hexaware Technologies

July 2024 - May 2025

- Utilized **SQL** and **Python** to query, **analyze**, and transform large datasets, generating insights for troubleshooting and performance optimization.
- Developed **automation** scripts to streamline backend processes and improve data accuracy and efficiency.
- Collaborated with business and technical teams to translate requirements into **data-driven** reports and solutions.

INTERNSHIP

Data Analyst, Center of business analytics

Nov 2025 - Present

- Designed the **relational** database for an internal portal, including **ER diagrams**, table structures, and **relationship** design to ensure scalable and consistent data storage.
- Developed backend functionality in Python to handle data ingestion, **validation**, and business-rule processing, integrating the application with the **database**.
- Used **Python (pandas, NumPy, etc.)** to clean and analyze operational data, producing insights that informed portal features and performance improvements.

Backend Developer, Amazing Humanbeing

Feb 2022 - Jan 2023

- During my internship, I learned and developed backend features using **Node.js**, **Express.js**, and **MySQL**, including a location-based user search powered by **optimized SQL queries**.
- We designed and managed database relationships, CRUD operations, and data imports (**Excel/CSV** → **MySQL**) to support scalable application features.
- Implemented **JWT authentication** and authorization, ensuring secure and efficient user access.

TECHNICAL SKILLS

- Languages & Libraries:** Python (pandas, NumPy, scikit-learn), R, SQL, JavaScript, Node.js
- Databases & Cloud:** MySQL, AWS Aurora, Relational DB design, ERD, Normalization, AWS Bedrock
- Analytics & BI:** Tableau, Power BI (DAX, Power Query), Excel (Solver, Advanced), Flask
- Machine Learning:** Logistic Regression, Random Forest, KNN, SVM, LDA, NLP, Sentiment Analysis
- Time Series & Stats:** VAR, ARIMA, Granger Causality, IRF, FEVD, Regression, Hypothesis Testing
- Other:** Data Migration, ETL, Web Scraping, Linear Programming, Sensitivity Analysis, Git

ACADEMIC PROJECTS

VAR Analysis of U.S. Monetary Policy (R, Time Series)

Jan 2026 – Mar 2026

- Modeled Unemployment, Federal Funds Rate, and Inflation using VAR on 752 monthly observations (1954–2017); confirmed all three monetary channels via Granger causality ($p < 0.001$).
- Applied IRF and FEVD to show policy effects build over 6–18 months, with cross-variable influences explaining 30–40% of forecast uncertainty at 24 months.

Diabetes Risk Prediction (Logistic Regression, KNN, Random Forest, SVM)

Jan 2026 – Mar 2026

- Built a classification model on 253K+ CDC BRFSS records using 21 health indicators; optimized for at-risk recall as the primary clinical metric and selected Random Forest as the final model.
- Conducted threshold tuning, worst-case error analysis, and age-group robustness checks; RF achieved the best balance of recall, accuracy, and interpretable variable importance.

Supply Chain Logistics Optimization (Excel, Linear Programming)

Jan 2026 – Mar 2026

- Developed an LP optimization model for a 19-plant, 10-port supply chain minimizing daily logistics cost; used Excel Solver and performed bottleneck/sensitivity analysis (shadow prices).
- Identified a single-port bottleneck through scenario analysis; targeted investment recommendations delivered greater cost savings than broad network changes.

Uber vs Ola Reviews Comparison (R, Text Mining, Topic Modeling)

Oct 2025 - Dec 2025

- Analyzed 1.4M+ Google Play reviews for Uber and Ola in India, performing large-scale data cleaning, sentiment analysis, and keyword-based issue tagging (price, driver, cancellations, safety)
- Applied LDA topic modeling and logistic regression to uncover key complaint themes and quantify how sentiment, review length, and platform affect the probability of 4–5-star ratings and used **Power BI** to visualize it

IMDb Top 1000 Movies Analytics Project (Python, NLP, Regression)

Oct 2025 - Dec 2025

- Scraped and cleaned IMDb Top 1000 movies data, engineering features such as year, runtime, decade, main genre, votes, revenue, description length, and sentiment.
- Performed exploratory, bivariate, and multivariate analysis with visualizations(**Tableau**) to study popularity, genre trends, and critic vs. audience scores; built a regression model to predict revenue from ratings, votes, runtime, and text features.

CAMPUS & LEADERSHIP ACTIVITIES

Graduate Program Ambassador

Aug 2025 - Present

- Represent the university at graduate events, engaging with prospective students.
- Support recruitment, outreach, and student engagement initiatives.

Certifications

- SQL Advanced & Python Basic (HackerRank, 2025) • Generative AI with AWS Bedrock (UNH, 2025) • R Programming (Simplilearn, 2025) • Complete Python Bootcamp (Udemy, 2024) • Data Analytics Workshop (Guru Nanak College, 2022)

Additional activities:

- Workshop in data analytics, pre-placement workshop.